



Synthesis and Characterization of PVA–hematite nanocomposite film



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Level-Term : 04-01

Experiment

Number : B3

Group : A1

Course number: MME408

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Performance date: 09-07-2025

Submission date: 16-07-2025

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Introduction:

In the last several years, advances in nanotechnology have helped in developing multifunctional polymer-based nanocomposites with enhanced performance properties. Of the numerous, polyvinyl alcohol (PVA) has been considered a promising matrix material due to its biodegradability, biocompatibility, film-forming ability, and solubility in water. These characteristics make PVA a green and industrially versatile polymer. PVA composites, upon being combined with nanomaterials such as nanoparticles, nanotubes, or nanofibers, exhibit elevated mechanical, thermal, and barrier properties.

One such innovation involves incorporating hematite (Fe_2O_3) nanoparticles into the PVA matrix to create PVA–hematite nanocomposite films. Incorporation of hematite imparts magnetic functionality and structural reinforcement, enhancing the range of potential applications of the composite in biomedical devices, packaging, electronics, and sensor technology. Such composites are usually developed via solution casting and sol-gel techniques, which allow for uniformity and controlled thickness of the resulting films.

The addition of Fe_2O_3 into PVA not only maintains the beneficial properties of PVA but also creates an impartation of new functions that increase its range of applications in technological and biomedical industries.

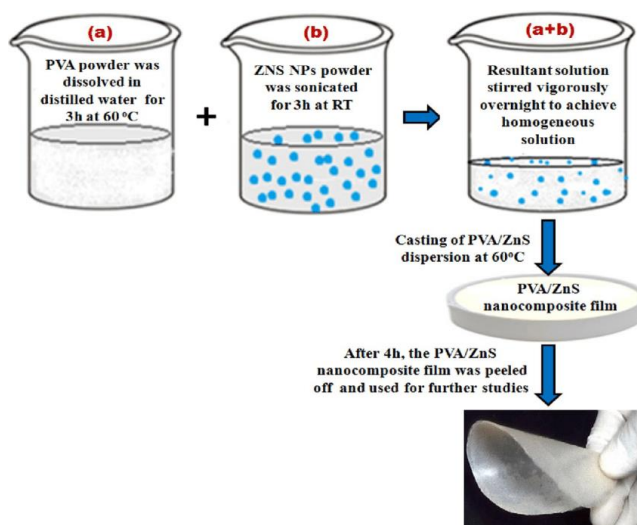


Figure 1: Example of PVA-nanocomposite film

Objectives:

1. To synthesize PVA–hematite nanocomposite films by dispersing Fe_2O_3 nanoparticles in a polyvinyl alcohol matrix using the solution casting method.
2. To understand and execute the solution casting process consisting of polymer dissolution, dispersion of the nanoparticles, casting, and controlled drying to achieve thin films.
3. To study the PVA– Fe_2O_3 nanoparticle interaction by analyzing the structure and chemical properties of the resultant film using Fourier Transform Infrared Spectroscopy (FTIR).
4. To examine the impact of nanoparticle incorporation on chemical bonding and potential change in functional groups within the polymer matrix.
5. To determine the applications of PVA-based nanocomposite synthesis for actual uses in packaging, biomedical, or electronic applications due to their improved features.

Equipment:

- PVA(Polyvinyl alcohol)
- DI water
- Magnetic stirrer
- Fe_2O_3 nanoparticles
- Mortar
- Mould
- FTIR machine

Experimental Procedures:

- I. 10 grams of polyvinyl alcohol (PVA) were weighed and added slowly to 100 mL of deionized water in a beaker.
- II. The beaker was placed on a hot plate and heated to 70°C with constant stirring of the solution until the PVA was completely dissolved and a homogenous solution was achieved.

- III. After complete dissolution, **0.1 grams of Fe_2O_3** (hematite) nanoparticles were added to the PVA solution. It was stirred thoroughly to ensure uniform distribution of the nanoparticles within the polymer matrix.
- IV. The solution was allowed to stand for a brief moment to eliminate air bubbles that were formed during the stirring process. It was then cast slowly into a clean mold or Petri dish to form a uniform layer.
- V. The cast film was dried in air at room temperature to allow complete evaporation of the solvent, resulting in the formation of a solid PVA- Fe_2O_3 nanocomposite film.
- VI. The film, once dried, was cautiously withdrawn from the mold and characterized by Fourier Transform Infrared Spectroscopy (FTIR) for chemical interactions and to ensure the presence of hematite nanoparticles.



Figure 3: Fe_2O_3 Nanoparticles



Figure 2: Hematite NPs in furnace after sintering



Figure 4: Adding hematite NPs in PVA solution

Data and Graph:

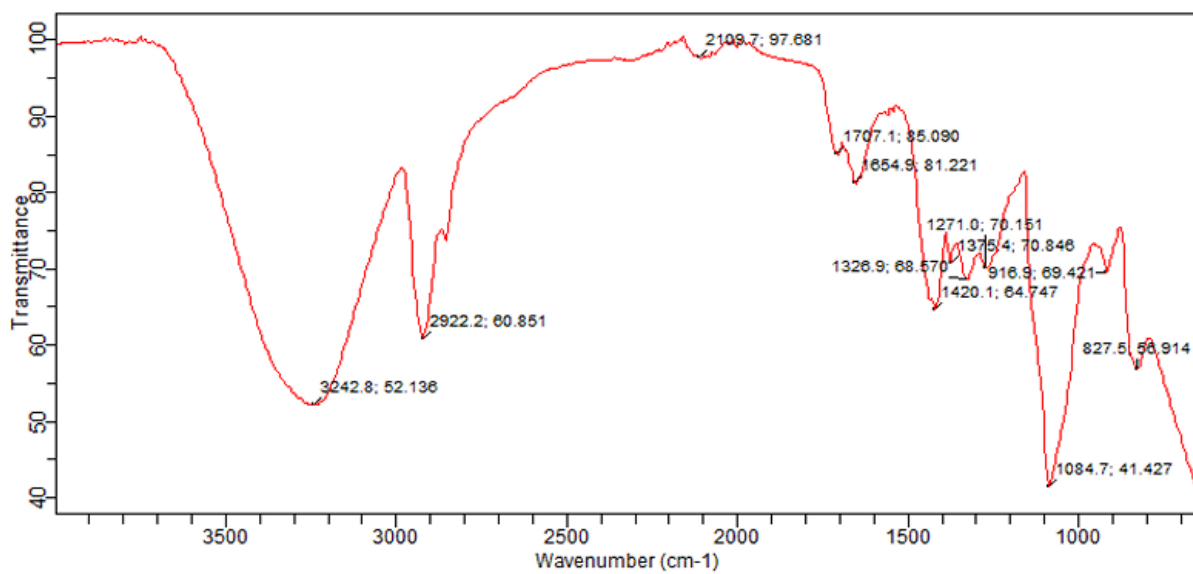


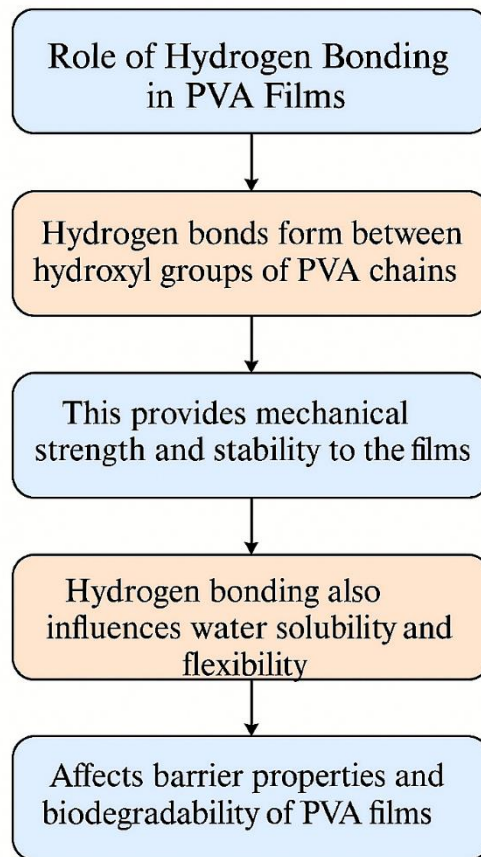
Figure 5: FTIR curve of our product film

Discussion:

1) What role does hydrogen bonding play in PVA films?

Hydrogen bonding plays a crucial role in defining the structural and functional properties of PVA films.

- **Inter-molecular Cohesion:** Hydrogen bonds form between hydroxyl (-OH) groups in adjacent PVA chains, giving the film mechanical strength and structural strength.
- **Solubility and Swelling:** They also form hydrogen bonds with water molecules, thereby making PVA water soluble and swellable.
- **Barrier Properties:** PVA films possess low permeability due to extensive hydrogen bonding, thus increasing barrier efficiency.
- **Flexibility and Biodegradability:** Hydrogen bonding provides a balance between rigidity and flexibility. [1][2]



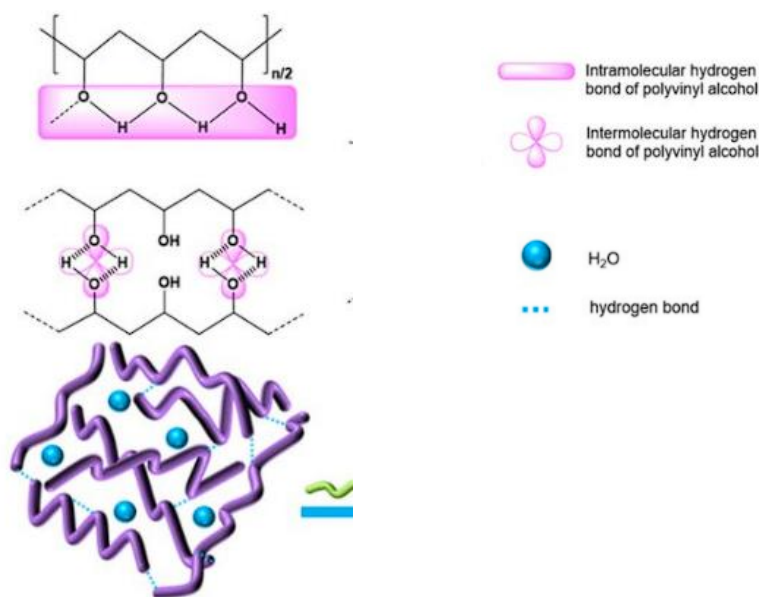


Figure 6: Intermolecular H₂ bond of PVA

2) Compare the FTIR results of your nanocomposite film with those of the pristine film and identify which bonds of the polymer have been affected by interaction with the nanoparticles.

In order to understand the molecular-level changes induced due to the addition of hematite (Fe₂O₃) nanoparticles into the PVA matrix, comparative FTIR (Fourier Transform Infrared Spectroscopy) study was performed between the pure PVA film and PVA–hematite nanocomposite film.

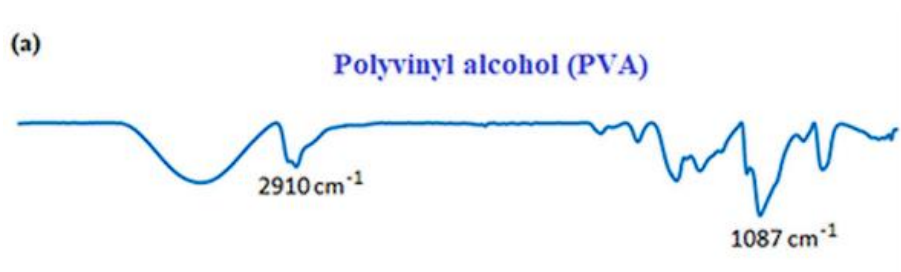
The pure PVA film showed typical absorption bands at $\sim 3242\text{ cm}^{-1}$, corresponding to the O–H stretching mode of strong intermolecular hydrogen bonding in hydroxyl groups. Peaks at $\sim 2922\text{ cm}^{-1}$ and $\sim 1141\text{ cm}^{-1}$ were due to C–H stretching and C–O stretching, respectively—typical features of the polymer chain.

Upon addition of Fe₂O₃ nanoparticles, typical spectral shifts were observed:

- The broad, wide O–H peak widened and shifted as a consequence of altered hydrogen-bonded environments. Peaks towards lower wavenumbers (about 3200–3400 cm^{-1}) indicate stronger or more advanced hydrogen-bonding interactions between PVA hydroxyl groups and nanoparticle surfaces.
- Peaks appearing near the $\sim 1700 \text{ cm}^{-1}$ region represented probable C=O group interactions, whether due to newly formed carbonyl species or coordination related to nanoparticles.
- Small changes in C–H and C–O regions ($\sim 2900\text{--}1000 \text{ cm}^{-1}$) reflected conformational changes of the PVA chain due to nanoparticle incorporation.

These spectral variations are in agreement with the hypothesis that Fe_2O_3 nanoparticles physically as well as chemically interact with PVA functional groups, disrupting native hydrogen bonding to form new networks of hydrogen bonds or coordination geometries. These interactions are essential in explaining the improved mechanical strength, thermal stability, and barrier properties of nanocomposite films.

Moreover, The FTIR spectrum identifies the sample as predominantly polyvinyl alcohol (PVA) of high hydrolysis content ($\sim 99.7\%$) and molecular weight $\sim 115,000$, based on a good spectral similarity (similarity index 0.90468). Common O–H stretching bands near 3200–3400 cm^{-1} identify the presence of strong hydrogen bonding typical of PVA. The spectrum further indicates weak similarities with ethylene-vinyl alcohol copolymer and glycerol suggestive of potential plasticizers or copolymer residues. No intense new peaks were observed that would indicate chemical bonding with external nanoparticles, and the sample must therefore either be pure PVA film or a nanocomposite with physical (but not chemical) interactions. This will serve as a control to compare the peak shifts, broadenings, or intensity changes of nanocomposite films representing hydrogen bonding or coordination with nanoparticles.[3]



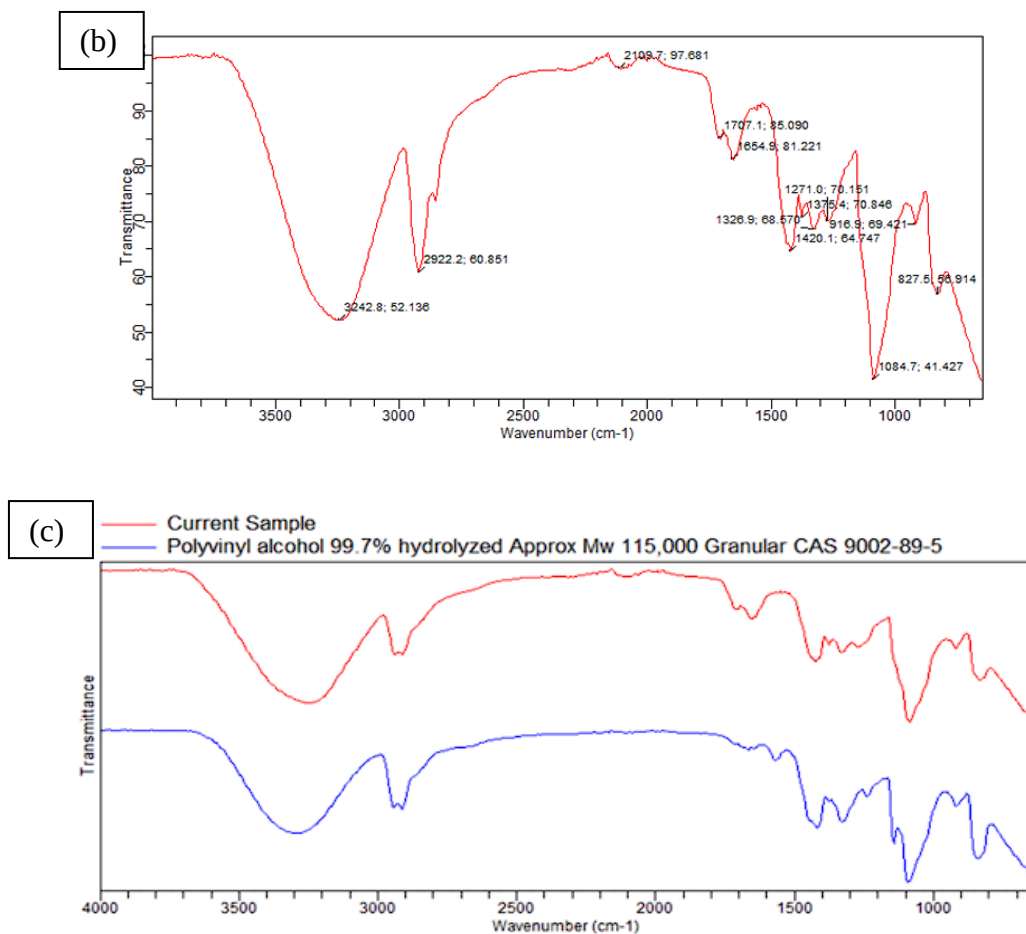


Figure 7: FTIR spectrum of a)Pristine PVA(from paper) b) My sample c)Comparison of Pristine PVA with our product (from provided pdf file)

FTIR peak shift between pure PVA film and nanocomposite film is due to interactions of Fe₂O₃ nanoparticles with polymer chains. The nanoparticles form hydrogen bonding with hydroxyl groups of PVA, specifically affecting the O–H stretching region (~3300 cm⁻¹), which can shift or broaden. Lower shifts in the C–O and C–H regions (~1100–2900 cm⁻¹) represent changes in electron distribution due to nanoparticle presence. New or enhanced peaks at 500–600 cm⁻¹ are signs of Fe–O vibrations, proving hematite incorporation. In summary, these shifts demonstrate strong chemical or physical interactions between hematite and PVA, changing molecular vibrations.

3)How can you reduce the water solubility of PVA films?

To reduce the water solubility of PVA films, some efficient methods are used. The most common one is crosslinking PVA chains with crosslinking agents like glutaraldehyde or borax, forming a tight 3D network to block water penetration. Another method is introducing hydrophobic components—either by blending with hydrophobic polymers or grafting alkyl/fluorine groups—to reduce water affinity. Addition of nanoparticles such as silica or clay also contributes film density and creates a physical barrier to water. Raising the molecular weight or varying the degree of hydrolysis of PVA also enhances water resistance by limiting hydrogen bonding sites. All these modifications make PVA films usable in applications where water resistance plays an important role, for example, packaging, biomedical films, and coatings.

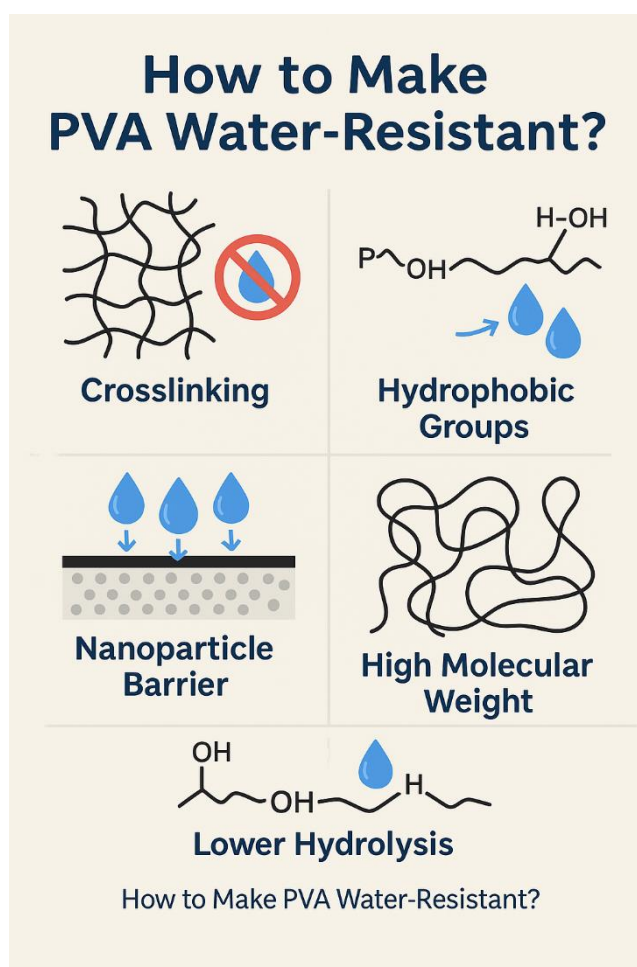


Figure 8: 5-ways to make PVA water resistant

References:

- [1] C. M. Hassan and N. A. Peppas, "Structure and applications of poly(vinyl alcohol) hydrogels produced by conventional crosslinking or by freezing/thawing methods," *Advances in Polymer Science*, vol. 153, pp. 37–65, 2000. doi: 10.1007/3-540-46414-X_2.
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